

VIDEO 4

OUTLIER DETECTION



BITS Pilani
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Probability And Statistics

Outlier Detection

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Introduction



In this video we will discuss

- ▶ The procedure to identify outliers in the given data



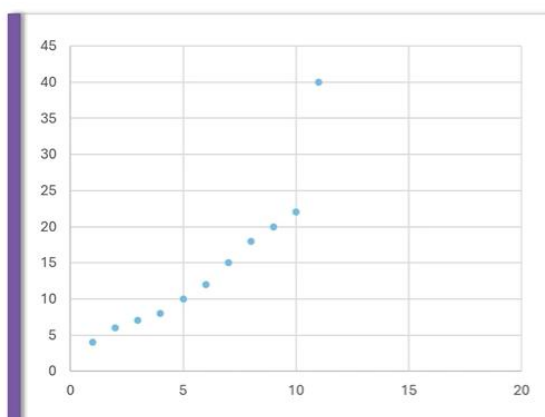
Outlier

An outlier in a data set is a data point that significantly deviates from the other values.

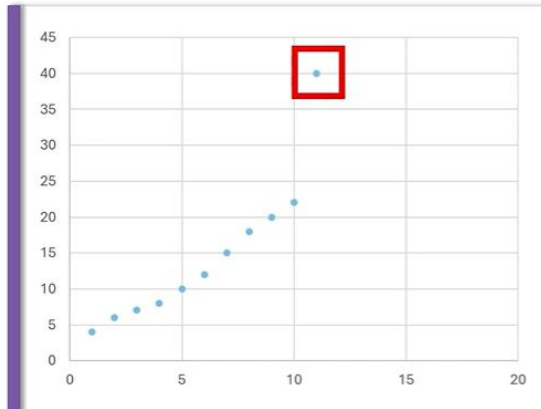
These extreme values can be much higher or lower than the majority of the data and may indicate unusual occurrences, errors, or genuine rare events.



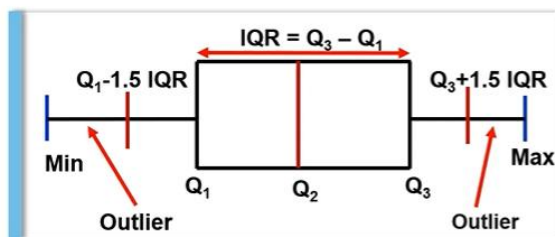
Outlier



Outlier



Outlier



Example

Find the outlier of the following
4,6,7,8,10,12,15,18,20,22,40

- First Quartile $Q_1 = 7$
- Third Quartile $Q_3 = 18$
- Inter Quartile Range(IQR) = $18 - 7 = 11$
- Lower Bound = $Q_1 - 1.5 \text{ IQR} = 7 - 1.5(11) = -9.5$
- Upper Bound = $Q_3 + 1.5 \text{ IQR} = 18 + 1.5(11) = 34.5$
- Any value < -9.5 or > 34.5 is an outlier
- $40 > 34.5$ is an outlier



Outlier

DATA 4,6,7,8,10,12,15,18,20,22,40

